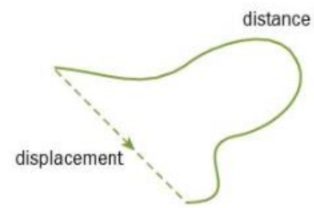
1. Vectors Gate — Vectors
    a. Vector notation

A **vector** is a quantity which has both magnitude (size) and direction.

The vector $\begin{pmatrix} a \\ b \end{pmatrix}$ can be used to describe a displacement of a units in the x-direction and b units in the y-direction.

A **scalar** is a quantity which has magnitude but no associated direction.

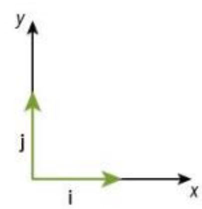
Distance is an example of a scalar.



In two dimensions a displacement can be represented as $\begin{pmatrix} x \\ y \end{pmatrix}$.

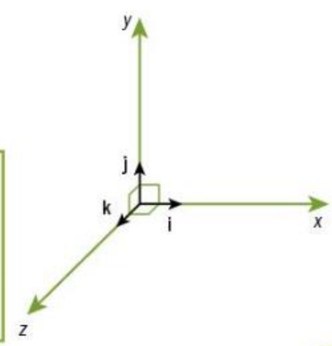
Positive and negative are used to denote directions as in the standard two-dimensional plane.

The **column vector** $\begin{pmatrix} x \\ y \end{pmatrix}$ can be represented in the form $x\mathbf{i} + y\mathbf{j}$, where $\mathbf{i}$ and $\mathbf{j}$ are **unit vectors** (vectors of length 1 unit) in the x and y directions respectively.



In three dimensions a displacement can be represented as $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$.

The diagram shows a three-dimensional set of axes, x, y, and z.



The column vector $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ can be represented as $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, where $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are unit vectors in the x, y, and z directions respectively.

Example 1

a) Write down the displacement from $A(2, 6)$ to $B(5, 7)$ as a column vector.

b) Write down the displacement from $P(2, 1, 3)$ to $Q(7, 2, -1)$ in terms of the unit vectors **i**, **j**, **k**.

a) $\overrightarrow{AB} = \begin{pmatrix} 5-2 \\ 7-6 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$

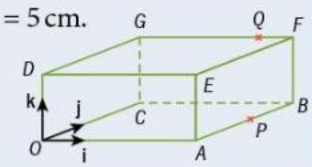
Show the increase in both coordinates as two positive components.

b) $\overrightarrow{PQ} = (7-2)\mathbf{i} + (2-1)\mathbf{j} + (-1-3)\mathbf{k} = 5\mathbf{i} + \mathbf{j} - 4\mathbf{k}$

Show the decrease in the z-coordinate by using a negative component.

Example 2

In the diagram, $OABCDEFG$ is a cuboid with $OA = 10$ cm, $AB = 6$ cm, $OD = 5$ cm. The unit vectors **i**, **j**, **k** are in the directions OA, OC, OD respectively. The point P is the mid-point of AB and Q lies on GF such that $GQ = 4QF$. Express each of the vectors $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ in terms of **i**, **j**, **k**.



$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = 10\mathbf{i} + 3\mathbf{j} + 0\mathbf{k} = 10\mathbf{i} + 3\mathbf{j}$

10 units in x-direction, 3 units in y-direction and 0 units in z-direction

$\overrightarrow{OQ} = 8\mathbf{i} + 6\mathbf{j} + 5\mathbf{k}$

$GQ = \frac{4}{5} GF = \frac{4}{5} \times 10 = 8$

Example 3

If A has coordinates $A(3, -2, 4)$ and B has coordinates $B(5, -3, 2)$, find as column vectors

a) $\overrightarrow{AB}$ **b)** $\overrightarrow{BA}$.

a) $\overrightarrow{AB} = \begin{pmatrix} 5-3 \\ -3--2 \\ 2-4 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix}$

More commonly we think of $\overrightarrow{AB}$ as being the coordinates of A subtracted from the coordinates of B.

b) $\overrightarrow{BA} = \begin{pmatrix} 3-5 \\ -2--3 \\ 4-2 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$

Subtract the coordinates of B from the coordinates of A.

b. Magnitude of vectors

Example 5

a) Find the magnitude of the vector $\begin{pmatrix} 3 \\ 4 \\ 6 \end{pmatrix}$.

b) Hence write down a unit vector in the direction of $\begin{pmatrix} 3 \\ 4 \\ 6 \end{pmatrix}$.

a) Magnitude of $\begin{pmatrix} 3 \\ 4 \\ 6 \end{pmatrix} = \sqrt{3^2 + 4^2 + 6^2} = \sqrt{61}$

Find the length of the vector.

b) Unit vector $= \frac{1}{\sqrt{61}} \begin{pmatrix} 3 \\ 4 \\ 6 \end{pmatrix}$

This is equivalent to dividing each component of the vector by $\sqrt{61}$.

Example 6

a) Find the value of k for which the vectors $\begin{pmatrix} 2 \\ -1 \\ 6 \end{pmatrix}$ and $\begin{pmatrix} 8 \\ -4 \\ k \end{pmatrix}$ are parallel.

b) Find the value of t for which the vectors $\begin{pmatrix} 5+t \\ 4 \\ 3-3t \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 4 \\ 9 \end{pmatrix}$ are equal.

a) For the vectors to be parallel, they must be scalar multiples of each other.

$\begin{pmatrix} 8 \\ -4 \\ k \end{pmatrix} = 4 \begin{pmatrix} 2 \\ -1 \\ 6 \end{pmatrix}$

Find the scalar multiple.

$k = 4 \times 6 = 24$

Use the third component to find k.

b) For the vectors to be equal, all three components must be equal.

$5 + t = 3$ and $3 - 3t = 9$

$t = -2$

Equate each component.

c. Geometric approach to addition/subtraction of vectors

Example 7

a) A is a point with coordinates $(1, 0, 3)$ and B is a point with coordinates $(3, -2, 6)$.
Find, using column vectors,

i) $\overrightarrow{AB}$

ii) the position vector of the point C, which is the mid-point of AB.

b) $OADB$ is a parallelogram.

Find the position vector of the point D.

c) What do your answers to **(a)** and **(b)** tell you?

a) i) $\overrightarrow{AB} = \begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$

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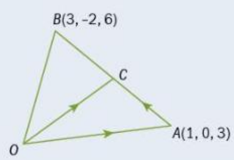
ii) $\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC}$

but $\overrightarrow{AC} = \frac{1}{2}\overrightarrow{AB} = \begin{pmatrix} 1 \\ -1 \\ \frac{3}{2} \end{pmatrix}$

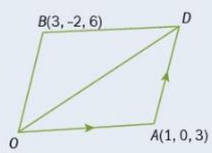
so $\overrightarrow{OC} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \\ \frac{3}{2} \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ \frac{9}{2} \end{pmatrix}$



b) $\overrightarrow{OD} = \overrightarrow{OA} + \overrightarrow{AD}$

$= \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \\ 6 \end{pmatrix}$

$\overrightarrow{AD} = \overrightarrow{OB}$



$\overrightarrow{OD} = \begin{pmatrix} 4 \\ -2 \\ 9 \end{pmatrix}$

c) $\overrightarrow{OD} = 2\overrightarrow{OC}$ so C is the mid-point of OD.

Therefore the diagonals of the parallelogram bisect each other.

Example 8

The points A, B, and C are such that
$\overrightarrow{OA} = 2\mathbf{i} + 6\mathbf{j} + 3\mathbf{k}$, $\overrightarrow{OB} = \mathbf{i} + 2\mathbf{j} + 7\mathbf{k}$, and $\overrightarrow{OC} = 4\mathbf{i} + 14\mathbf{j} - 5\mathbf{k}$.

Show that the vectors $\overrightarrow{BA}$ and $\overrightarrow{AC}$ are in the same direction and hence that A, B, C lie on the same straight line.

$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB}$

$= (2\mathbf{i} + 6\mathbf{j} + 3\mathbf{k}) - (\mathbf{i} + 2\mathbf{j} + 7\mathbf{k})$

$\overrightarrow{BA}$ = position vector of A minus position vector of B

$= \mathbf{i} + 4\mathbf{j} - 4\mathbf{k}$

$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA}$

$= (4\mathbf{i} + 14\mathbf{j} - 5\mathbf{k}) - (2\mathbf{i} + 6\mathbf{j} + 3\mathbf{k})$

$\overrightarrow{AC}$ = position vector of C minus position vector of A

$= 2\mathbf{i} + 8\mathbf{j} - 8\mathbf{k}$

$\overrightarrow{AC} = 2\overrightarrow{BA}$ so $\overrightarrow{BA}$ and $\overrightarrow{AC}$ are parallel.

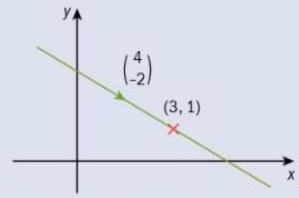
But A lies on both $\overrightarrow{BA}$ and $\overrightarrow{AC}$, so A, B, C lie on a straight line.

d. Vector equation of straight line

Example 9

a) Write down a vector equation for the straight line through the point (3, 1) with direction vector $\begin{pmatrix} 4 \\ -2 \end{pmatrix}$.

b) Find a vector equation for the line through the points $L(1, 2)$ and $M(5, 3)$.



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a) $\mathbf{a} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ — This is the position vector of a point on the line.

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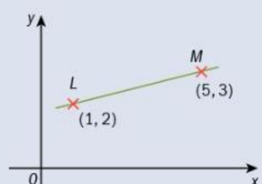
$\mathbf{b} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$ — This gives the direction of the line.

A vector equation of the line is therefore $\mathbf{r} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} + t\begin{pmatrix} 4 \\ -2 \end{pmatrix}$

b) $\mathbf{a} = \overrightarrow{OL} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ — Alternatively we could use $\overrightarrow{OM} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}$.

$\mathbf{b} = \overrightarrow{LM} = \begin{pmatrix} 5 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$ — Alternatively we could use $\overrightarrow{ML} = \begin{pmatrix} -4 \\ -1 \end{pmatrix}$ or any multiple of $\overrightarrow{LM}$ or $\overrightarrow{ML}$.



A vector equation of the line is therefore $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + t\begin{pmatrix} 4 \\ 1 \end{pmatrix}$

Remember: Include '**r** =' as part of the equation.

We can also use vectors to write the equation of a line in three dimensions, as Example 10 shows.

Example 10

Points A and B have position vectors $\begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 5 \\ 0 \end{pmatrix}$ respectively.

a) Find, in vector form, an equation of the straight line that passes through A and B.

b) Show that the point $C(-12, 13, 8)$ lies on this line.

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a) $\overrightarrow{AB} = \begin{pmatrix} -2 \\ 5 \\ 0 \end{pmatrix} - \begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix} = \begin{pmatrix} -5 \\ 4 \\ 4 \end{pmatrix}$ — $\overrightarrow{AB}$ = position vector of B minus position vector of A

A vector equation of the line is $\mathbf{r} = \begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix} + t\begin{pmatrix} -5 \\ 4 \\ 4 \end{pmatrix}$

b) $\begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix} + t\begin{pmatrix} -5 \\ 4 \\ 4 \end{pmatrix} = \begin{pmatrix} -12 \\ 13 \\ 8 \end{pmatrix}$ — For C to lie on the line there must be a value of t which, when substituted into the equation, gives the position vector of C.

Example 11

A line has a vector equation given by $\mathbf{r} = (3\mathbf{i} + \mathbf{j} + \mathbf{k}) + \mu(2\mathbf{i} - \mathbf{j} - \mathbf{k})$.

a) Show that this line intersects the x-axis.

b) The point P lies on this line and has coordinates $(-5, a, b)$. Find the values of a and b.

a) If the line intersects the x-axis then there will be a value of μ which gives a position vector of the form $c\mathbf{i} + 0\mathbf{j} + 0\mathbf{k}$.

Any point on the x-axis has position vector with y and z components equal to zero.

$(3\mathbf{i} + \mathbf{j} + \mathbf{k}) + \mu(2\mathbf{i} - \mathbf{j} - \mathbf{k}) = c\mathbf{i} + 0\mathbf{j} + 0\mathbf{k}$

$(3 + 2\mu)\mathbf{i} + (1 - \mu)\mathbf{j} + (1 - \mu)\mathbf{k} = c\mathbf{i} - 0\mathbf{j} - 0\mathbf{k}$

Rewrite the equation so we can compare the **i**, **j**, and **k** components.

$3 + 2\mu = c$

Compare the **i** components.

$1 - \mu = 0$ so $\mu = 1$

Compare the **j** and **k** components.

$\Rightarrow 3 + 2 \times 1 = c$

$\Rightarrow c = 5$

Substitute $\mu = 1$ into $3 + 2\mu = c$ to find the value of c.

When $\mu = 1$ and $c = 5$, $\mathbf{r} = 5\mathbf{i} - 0\mathbf{j} - 0\mathbf{k}$

so the point $(5, 0, 0)$ lies on the line and on the x-axis.

Substitute $\mu = 1$ and $c = 5$ into the equation of the line.

Therefore the line intersects the x-axis.

b) $(3\mathbf{i} + \mathbf{j} + \mathbf{k}) + \mu(2\mathbf{i} - \mathbf{j} - \mathbf{k}) = -5\mathbf{i} + a\mathbf{j} + b\mathbf{k}$

Equate the RHS of the equation of the line with the position vector of P.

$(3 + 2\mu)\mathbf{i} + (1 - \mu)\mathbf{j} + (1 - \mu)\mathbf{k} = -5\mathbf{i} + a\mathbf{j} + b\mathbf{k}$

$3 + 2\mu = -5$

$\mu = -4$

Collect terms in **i**, **j**, and **k** and compare components to find the value of μ.

When $\mu = -4$,

$\mathbf{r} = (3\mathbf{i} + \mathbf{j} + \mathbf{k}) + (-4)(2\mathbf{i} - \mathbf{j} - \mathbf{k})$

$= -5\mathbf{i} + 5\mathbf{j} + 5\mathbf{k}$

Substitute $\mu = -4$ into the equation of the line.

$a = 5, b = 5$

State the values of a and b.

e. Intersecting/parallel/skew lines

Example 12

Two lines l_1 and l_2 are given by the vector equations

$\mathbf{r} = \begin{pmatrix} 5 \\ 1 \\ -1 \end{pmatrix} + t\begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}$ and $\mathbf{r} = \begin{pmatrix} 13 \\ -6 \\ 2 \end{pmatrix} + \mu\begin{pmatrix} -3 \\ 4 \\ 1 \end{pmatrix}$ respectively.

Show that the lines intersect and find the position vector of the point of intersection.

We can see that either the lines must intersect, or else they are skew. Their directions, given by $\begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} -3 \\ 4 \\ 1 \end{pmatrix}$, are not the same and are not multiples of each other, so the lines are not parallel or coincident.

$$\begin{pmatrix} 5 \\ 1 \\ -1 \end{pmatrix} + t\begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} = \begin{pmatrix} 13 \\ -6 \\ 2 \end{pmatrix} + \mu\begin{pmatrix} -3 \\ 4 \\ 1 \end{pmatrix}$$

For the lines to intersect there must be a value of t and a value of μ which give the same position vector from each equation.

$$5 + 2t = 13 - 3\mu \qquad (1)$$
$$1 + t = -6 + 4\mu \qquad (2)$$
$$-1 + 5t = 2 + \mu \qquad (3)$$

Equation (1) – 2 × equation (2) gives

Solve two of the equations to find a pair of values which satisfies those two equations.

$$3 = 25 - 11\mu$$
$$\mu = 2$$

Substituting $\mu = 2$ into equation (2):

$$1 + t = -6 + 4 \times 2$$
$$t = 1$$

Checking equation (3):

Check that these values satisfy the third equation.

When $t = 1$ and $\mu = 2$, $-1 + 5t = 4$

$$2 + \mu = 4$$

So this pair of values also satisfies the third equation and we have shown that the lines intersect.

If we solve two of the equations to get a value for each of the scalars but then, on checking, these values do not satisfy the third equation, then the lines do not intersect.

$$\mathbf{r} = \begin{pmatrix} 5 \\ 1 \\ -1 \end{pmatrix} + 1\begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} = \begin{pmatrix} 7 \\ 2 \\ 4 \end{pmatrix}$$

Find the position vector of the point of intersection by substituting the value of t into the equation of l_1 (or alternatively the value of μ into the equation of l_2).

If we were asked to find the **coordinates** of the point of intersection, we would give the answer in the form (7, 2, 4).

Example 13

Determine whether the following pairs of lines intersect. If they do not intersect, determine whether they are parallel or skew.

a) $\mathbf{r} = \begin{pmatrix} 4 \\ 5 \\ 2 \end{pmatrix} + t\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}, \quad \mathbf{r} = \begin{pmatrix} 5 \\ 0 \\ 4 \end{pmatrix} + \mu\begin{pmatrix} -4 \\ 2 \\ -6 \end{pmatrix}$

b) $\mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ 1 \end{pmatrix} + t\begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}, \quad \mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix} + \mu\begin{pmatrix} 5 \\ -2 \\ 4 \end{pmatrix}$

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a) By inspection we see that

$$\begin{pmatrix} -4 \\ 2 \\ -6 \end{pmatrix} = -2\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$

The lines are therefore parallel and do not intersect.

Firstly check to see if the lines are parallel.

It is easy to check whether the lines are parallel by looking at the direction vectors, so it can save work to do this first.

b) The lines are not parallel.

Check the direction vectors in order to conclude this.

For intersection

$4 + 3t = 2 + 5\mu \qquad (1)$

$3 + 1t = 1 - 2\mu \qquad (2)$

$1 + 2t = 6 + 4\mu \qquad (3)$

Equation (1) – 3 × equation (2) gives

$-5 = -1 + 11\mu$

$\mu = -\frac{4}{11}$

Solve two of the equations for t and μ values.

Substituting $\mu = -\frac{4}{11}$ into equation (1) :

$4 + 3t = 2 + 5 \times -\frac{4}{11}$

$t = -\frac{14}{11}$

Checking equation (3):

$1 + 2t = 1 + 2 \times -\frac{14}{11} = -\frac{17}{11}$

$6 + 4\mu = 6 + 4 \times -\frac{4}{11} = \frac{50}{11}$

$1 + 2t \neq 6 + 4\mu$

Check to see if the values found also satisfy the third equation.

There are no values which satisfy all three equations, so there is no point in common for these two lines.

The lines are not parallel so the lines must be skew.

f. Scalar products

Example 14

Vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are given by $\overrightarrow{OA} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 6 \\ -4 \\ -5 \end{pmatrix}$.

Calculate the scalar product $\overrightarrow{OA} \cdot \overrightarrow{OB}$.

$$\overrightarrow{OA} \cdot \overrightarrow{OB} = (3 \times 6) + (-1 \times -4) + (2 \times -5)$$
$$= 12$$

Using $a_1b_1 + a_2b_2 + a_3b_3$

Example 15

The position vectors of points A and B relative to an origin O are given by $\mathbf{a} = 2\mathbf{i} + p\mathbf{j} + 3\mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} - \mathbf{j} + p\mathbf{k}$, where p is a constant.
Calculate $\mathbf{a}\cdot\mathbf{b}$.

$$\mathbf{a}\cdot\mathbf{b} = (2 \times 3) + (p \times -1) + (3 \times p)$$
$$= 2p + 6$$

Using $a_1b_1 + a_2b_2 + a_3b_3$

g. Angle between two lines

Example 16

Find the angle between the two vectors $\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}$.

$$\cos\theta = \frac{a_1b_1 + a_2b_2 + a_3b_3}{|\mathbf{a}||\mathbf{b}|}$$

Use the standard result.

$$= \frac{(1\times 2)+(-1\times 1)+(1\times 4)}{\sqrt{1^2+(-1)^2+1^2}\sqrt{2^2+1^2+4^2}}$$

Substitute the values.

$$= \frac{5}{\sqrt{3}\sqrt{21}}$$

$\theta = 51.0°$ (1 d.p.)

Give angle answers in degrees to 1 decimal place.

Example 17

a) Find the value of p if the vectors $p\mathbf{i} - 3\mathbf{k}$ and $2\mathbf{i} + \mathbf{j} + 5\mathbf{k}$ are perpendicular.

b) Show that the vectors $q\mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{i} + (q + 1)\mathbf{j} + \mathbf{k}$ are perpendicular.

a) The vectors are perpendicular if the scalar product = 0

$a_1b_1 + a_2b_2 + a_3b_3 = 0$

$(p \times 2) + (0 \times 1) + (-3 \times 5) = 0$

$2p - 15 = 0$

Simplify and solve the equation.

$p = \frac{15}{2}$

b) Scalar product $= (q \times 1) + (-1 \times (q + 1)) + (1 \times 1)$

Use $\mathbf{a}\cdot\mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$.

$= q - q - 1 + 1 = 0$

Therefore the vectors are perpendicular.

Write your conclusion.

Note: This means that the two vectors are perpendicular whatever the value of q is.

Example 18

Prove that the lines $\mathbf{r}_1 = (2t + 1)\mathbf{i} + (1 - t)\mathbf{j} + (1 - t)\mathbf{k}$ and $\mathbf{r}_2 = (3 + \mu)\mathbf{i} - (2\mu + 1)\mathbf{j} + (4\mu + 2)\mathbf{k}$ are perpendicular.

The equations of the lines can be written as

$\mathbf{r}_1 = \mathbf{i} + \mathbf{j} + \mathbf{k} + t(2\mathbf{i} - \mathbf{j} - \mathbf{k})$

$\mathbf{r}_2 = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k} + \mu(\mathbf{i} - 2\mathbf{j} + 4\mathbf{k})$

Write each equation in the form $\mathbf{r} = \mathbf{a} + t\mathbf{b}$.

But $(2\mathbf{i} - \mathbf{j} - \mathbf{k})\cdot(\mathbf{i} - 2\mathbf{j} + 4\mathbf{k})$

$= (2 \times 1) + (-1 \times -2) + (-1 \times 4)$

Find the scalar product of the direction vectors.

$= 2 + 2 - 4$

$= 0$

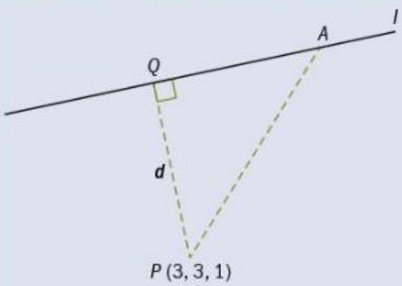
The scalar product is 0 so the lines are perpendicular.

Note: These two lines are perpendicular but happen to be skew.

h. Distance from point to line

Example 20

Find the distance of the point $P(3, 3, 1)$ from the line l whose equation is $\mathbf{r} = \begin{pmatrix} -4 \\ 3 \\ 0 \end{pmatrix} + t\begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix}$.



Any point on the line, say Q, has position vector

$$\mathbf{q} = \begin{pmatrix} -4+t \\ 3+2t \\ -4t \end{pmatrix}.$$

Find the general position vector of a point on the line.

The direction of l is $\begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix}$

For Q to be the foot of the perpendicular from P to the line,

$$\overrightarrow{PQ} \cdot \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} = 0$$

But $\overrightarrow{PQ} = \mathbf{q} - \mathbf{p} = \begin{pmatrix} -4+t \\ 3+2t \\ -4t \end{pmatrix} - \begin{pmatrix} 3 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} -7+t \\ 2t \\ -4t-1 \end{pmatrix}$

Find $\overrightarrow{PQ}$ in terms of t.

$$\begin{pmatrix} -7+t \\ 2t \\ -4t-1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} = 0$$

Substitute for $\overrightarrow{PQ}$ in $\overrightarrow{PQ} \cdot \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} = 0$.

$$(-7+t) \times 1 + (2t) \times 2 + (-4t-1) \times (-4) = 0$$
$$-7 + t + 4t + 16t + 4 = 0$$
$$t = \frac{1}{7}$$

Calculate the value of t.